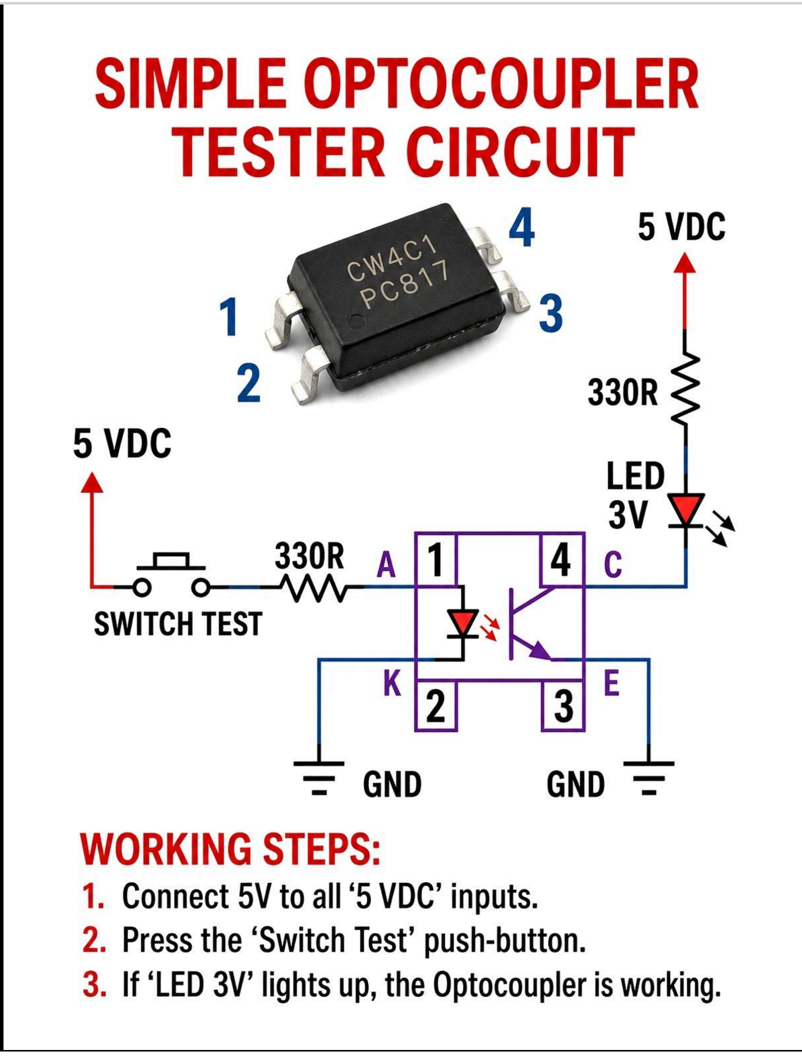

Electronics Education
5 maj kl. 16:30 ·

This circuit is a simple and practical way to test whether an optocoupler like the PC817 is working correctly. Inside the optocoupler, there are two completely isolated sections: an LED on the input side and a phototransistor on the output side. When you press the switch, 5 V is applied through a 330 Ω resistor to the internal LED (pins 1 and 2). This resistor limits the current and protects the LED. Once powered, the LED emits light inside the device. That light is detected by the phototransistor on the output side (pins 3 and 4), which then turns ON.

As the phototransistor conducts, current flows from the 5 V supply through another 330 Ω resistor and the external LED, then through the transistor to ground. This causes the external LED to glow, indicating that the optocoupler is functioning properly. The important concept here is electrical isolation—the input and output sides are not directly connected, and the signal is transferred using light. This makes optocouplers very useful for protecting sensitive circuits from noise, voltage spikes, or ground differences. If the LED does not turn ON during the test, it may indicate a faulty optocoupler, incorrect wiring, or insufficient input current.

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